

MacBook Wi-Fi dropping

Wi-Fi Connection Drops & Instability Troubleshooting

Sudden connection loss after established connections can be caused by network configuration conflicts, corrupted DHCP lease structures, or radio frequency interference from external hardware. Proceed through these steps sequentially to isolate and resolve.

Step 1: Renew the DHCP Lease

1. Go to **System Settings** (or *System Preferences*) > **Network**.
2. Select **Wi-Fi** and click the **Details** (or *Advanced*) button next to your active network connection.
3. Click **TCP/IP** in the left sidebar.
4. Click **Renew DHCP Lease**, then click **OK** and **Apply**.

Step 2: Create a Clean "Network Location"

Corrupted settings in older preference configurations can interfere with active leases. Forcing macOS to rebuild its networking profile often resolves continuous dropouts.

1. Within **Network** settings, locate the **Locations** dropdown menu (if hidden, look under the "... " or **More** menu icon).
2. Select **Edit Locations...** and click the plus **[+]** icon.
3. Name the new profile (e.g., **Home_New** or **Fix**) and click **Done**.
4. Select this newly created location profile from the dropdown menu, then click **Apply**.

Target path verified.

Step 3: Clear System Configuration Preference Files

If low-level system configuration files contain syntax errors or bad permissions, removing them forces macOS to rebuild clean configuration files automatically.

1. Turn your **Wi-Fi OFF**.
2. Open **Finder**, press **Cmd + Shift + G**.
3. Paste the following target path and press Enter:

```
/Library/Preferences/SystemConfiguration/
```

4. Select and delete (or move to a temporary folder on your desktop) the following files:

```
com.apple.airport.preferences.plist  
com.apple.network.eapolclient.configuration.plist  
com.apple.wifi.message-tracer.plist  
NetworkInterfaces.plist  
preferences.plist
```

5. **Restart** your Mac and toggle your **Wi-Fi back ON**.

Step 4: Isolate Hardware & Band Interference

- **Adjust Router Band Settings:** If your local network shares a single SSID across both 2.4GHz and 5GHz bands, the Mac may intermittently hop between them during signal drops. Try splitting the SSID names at your router level (e.g., **MyWiFi** vs. **MyWiFi_5G**) and pin the Mac directly to the stable 5GHz band.
- **Shield USB 3.0 & Hub Hardware:** Poorly shielded USB-C dongles, external drives, or uncertified displays emit noise on the 2.4GHz spectrum. Try disconnecting peripherals for 30 minutes to verify if performance stabilizes.

Step 5: Run Live Wireless Diagnostics

To analyze network drops, utilize the built-in diagnostic graphing tool:

1. Hold the **Option** key and click the **Wi-Fi icon** in your top menu bar.
2. Select **Open Wireless Diagnostics**.
3. Rather than running the setup wizard, go to the screen's top menu bar and select **Window > Performance**.
4. This exposes a live graph tracking signal strength (**RSSI**) and noise levels. A sudden spike in the "Noise" line correlating to the dropout confirms physical or environmental interference.